

Ruby & Ruby on Rails Training Curriculum

A structured topic list covering Ruby fundamentals, Ruby on Rails, and advanced engineering & testing practices.

Section	Topic Count
1. Ruby Fundamentals	27
2. Ruby on Rails	27
3. Advanced Topics	11
Total	65

1. Ruby Fundamentals

1. Installations of Softwares (RVM, Ruby, Rails, GIT, MySQL, Sublime Text Editor)
2. IRB Editor
3. Variables (Declaration and Access)
4. Number
5. Ranges
6. String
7. Symbol
8. Float
9. Loops
10. Flow Control
11. Scope Variables (Local, Global, Instance, Class)
12. Function / Methods (Instance and Class Method)
13. Hash
14. Array
15. Object Oriented Programming
16. Inheritance
17. Polymorphism
18. Encapsulation
19. Modules
20. Blocks
21. Closure (Procs and Lambdas)
22. Kernel Module
23. Files and Database
24. Exception Handling
25. Debugging
26. Dynamic Programming
27. Design Patterns

2. Ruby on Rails

1. Introduction
2. Rails Environment: Development, Staging, Production
3. MVC Architecture
4. Working with Controllers
5. Working with Models
6. Working with Views
7. Routing: REST, Resource
8. Scaffolding
9. HTML, HAML and ERB
10. Active Record (MySQL)
11. Active Record Migrations
12. Active Record Associations
13. Active Record Validation
14. Active Record Callbacks
15. Helper
16. Authentication and Authorization
17. AJAX on Rails including jQuery and JavaScript / StimulusJS
18. Bootstrap & CSS
19. Rails API
20. Gems & Plug-ins
21. Email Trigger
22. CSV & PDF Parser (Create / Read)
23. Image and File Upload
24. Thread
25. Concurrency
26. Garbage Collection
27. Caching

3. Advanced Topics

1. System Design
2. Managing Tasks with Rake (Ruby Make)
3. Elastic Search
4. Background Jobs / RabbitMQ
5. Test Driven Development
6. RSpec Introduction / Rails RSpec
7. Cucumber Introduction
8. Capybara Introduction
9. Test Doubles, Including Mocks and Stubs
10. Server Knowledge
11. Deployment